

cwltool

System	No Container	Singularity
LUMI	ERROR (3)	INVALID (29)
MareNostrum 5	OK (8)	ERROR (3)
Laptop	OK (3)	INVALID (4)
Cloud	OK (5)	INVALID (5)

toil

System	No Container	Singularity
LUMI	ERROR (124)	INVALID (113)
MareNostrum 5	ERROR (18)	ERROR (10)
Laptop	OK (11)	INVALID (14)
Cloud	OK (15)	INVALID (20)

Table 1: Simple MPI workflow using **baseCommand: mpirun**. Execution time is shown in seconds. “INVALID” indicates runs where the MPI job was launched via Singularity instead of the host MPI launcher.

cwltool

System	No Container	Singularity
LUMI	OK (4)	ERROR (4)
MareNostrum 5	OK (3)	ERROR (2)
Laptop	OK (3)	ERROR (3)
Cloud	OK (4)	ERROR (6)

toil

System	No Container	Singularity
LUMI	OK (14)	ERROR (20)
MareNostrum 5	ERROR (8)	ERROR (9)
Laptop	OK (11)	ERROR (14)
Cloud	OK (14)	ERROR (19)

Table 2: Simple MPI workflow using **MPIRequirement**. Execution time is shown in seconds.